

ELP305: Experiment No.2- Calculator

Objective: Designing a simple digital system using a micro controller.

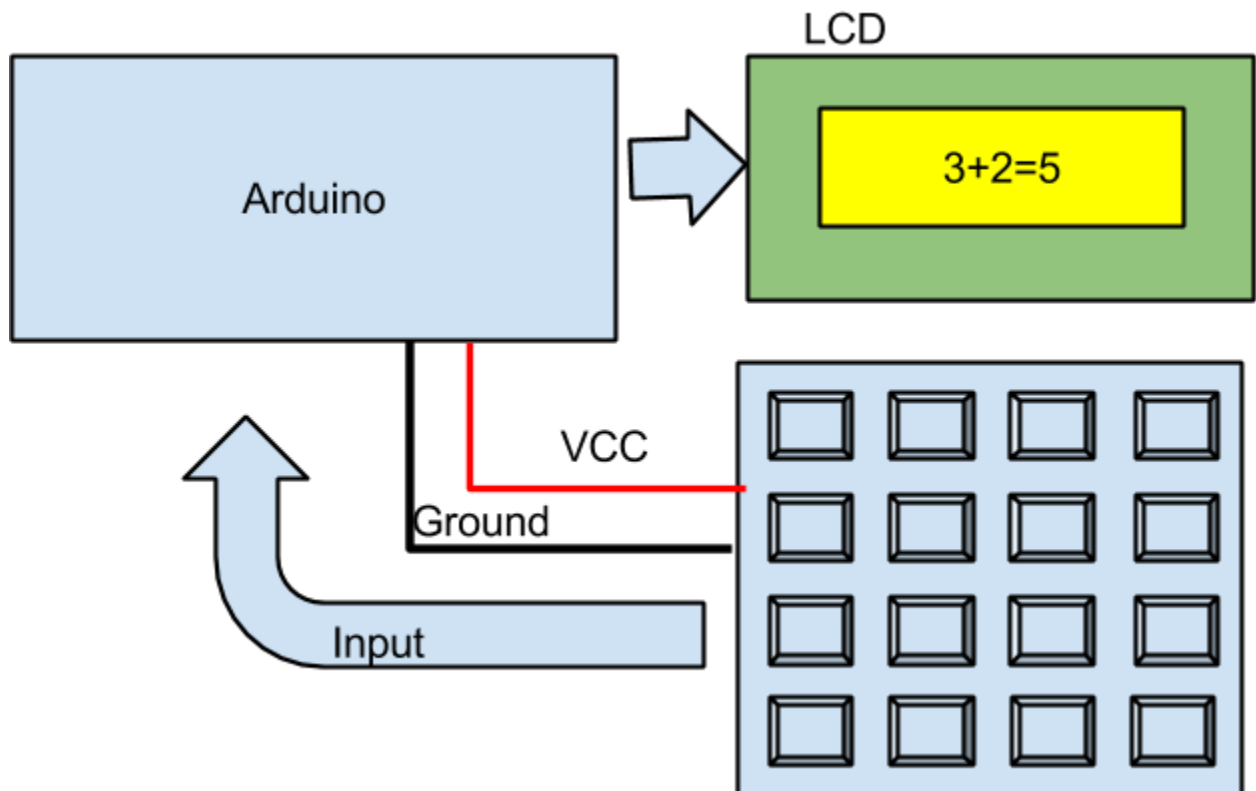
Problem Statement:

Make your own calculator. (Using arduino, 4 X 4 key pad and 2 X 16 LCD display).

Stuff you will need to have

1. Arduino
2. LCD
3. Keypad
4. breadboard
5. connecting wires

Schematic



Going Further

1 Extend this calculator to implement the SQRT() function.

1.a Does your calculator handle decimal nos. Or only integers? Can it handle a very large floating Point nos.? How large is the largest no. is? And implement the DMAS.

1.b How would you interface this calculator with a printer interface so that a hard-copy could be maintained of every transaction. (Obviously, we are not giving or making available a printer to you, So could you save all Transactions in a file with a time stamp which could be printed later?

2 To keep a record of what you have entered. You can see all the steps that you have done, in case you make a mistake somewhere. Turn ON the history function by pressing history button. The '+' and '-' key will help to move back and forth between steps.

2.a How would you incorporate parenthesis processing into your calculator.

2.b Would infix notation (what you are using now) be a worse choice than postfix or prefix notation?

3 Design a Digi-Lock which can be accessed through 3 different alpha-numeric passwords.

References

- 1 <http://playground.arduino.cc/Main/KeypadTutorial>